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How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

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Chapter 1: Exploring AI Concepts and Emerging Technologies

Artificial Intelligence (AI) is one of the fastest-growing fields of computer science. This chapter explains what AI is, how machines learn from data, how an AI system is built and why the data behind it matters so much.

Lesson	Title (MINESEC progression)
1	Introduction to artificial intelligence – Part 1
2	Introduction to artificial intelligence – Part 2
3	AI techniques and Machine Learning
4	Developing AI Systems and the importance of data in AI systems – Part 1
5	Developing AI Systems and the importance of data in AI systems – Part 2
6	Data collection and Data analysis
7	Integration Activity

Lesson 1: Introduction to artificial intelligence – Part 1

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✓ define artificial intelligence and explain why it has become practical
- ✓ classify AI by capability (narrow, general, super) and by function

KEY POINTS

Artificial Intelligence (AI) is the branch of computer science concerned with building systems (hardware and software) that perform tasks which normally require human intelligence: reasoning, learning, perception, problem solving and decision making.

In simple terms, AI is about making machines that can "think" and "learn" in ways that resemble human intelligence.

Why AI matters now

AI has existed as an idea since the 1950s, but it became practical in everyday life only in the last twenty years. Three factors made this possible:

- **Cheaper, more powerful computing:** modern processors and GPUs perform billions of calculations per second at low cost.
- **Huge amounts of digital data ("Big Data"):** phones, websites, sensors and transactions produce data every second, and AI needs data to learn.
- **Better algorithms:** especially deep learning, which has become far more effective since about 2012.

Categories of AI by capability

Category	Description	Example	Exists today?
Narrow (weak) AI	Designed for one specific task	Chess engine, spam filter, Google Translate	Yes: all AI in use today is narrow AI
General AI (AGI)	Human-level intelligence across any task	A machine that could learn and reason like a person in every field	No
Super AI	Intelligence far beyond the best humans in every field	Theoretical only	No

KEY POINTS

Despite what science fiction suggests, **all AI in use today is narrow AI**. Even the most advanced chatbots and self-driving cars are built for specific tasks and cannot generalise beyond them.

Categories of AI by function

Category	Description	Example
Reactive machines	Respond to the current input only; no memory of the past	IBM's Deep Blue (chess)
Limited memory	Use recent past data for a short time to make decisions	Self-driving cars
Theory of mind	Would understand emotions, beliefs and intentions (research stage)	None yet
Self-aware	Would have consciousness and a sense of self (theoretical)	None

Example: Classify each system by capability and by function: (a) a spam filter, (b) a self-driving car.

(a) Narrow AI; a reactive system (it classifies each email as it arrives).

(b) Narrow AI; limited memory (it uses the last few seconds of sensor data to decide).

Practice Questions: Lesson 1

1. State the three factors that made AI practical in recent years.
2. Classify a chess program as narrow, general or super AI, and as reactive or limited memory.

Answers: *1. Cheaper, more powerful computing; huge amounts of digital data; better algorithms (deep learning) 2. Narrow AI; reactive machine*

Lesson 2: Introduction to artificial intelligence – Part 2

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ identify everyday examples of AI and related emerging technologies
- ✔ discuss the benefits and concerns of AI

Everyday examples of AI

- **Voice assistants:** Siri, Alexa, Google Assistant.
- **Recommendation systems:** YouTube, TikTok, Netflix, online shops.
- **Facial recognition:** phone unlock, security cameras, passport control.
- **Spam filters** in email; **chatbots** for customer service.
- **Route optimisation:** Google Maps, Waze.
- **Medical diagnosis support:** analysing X-rays and scans.
- **Agriculture:** drones that monitor crops (precision farming).
- **Fraud detection** in banking and mobile money.
- **Machine translation:** English ↔ French with Google Translate or DeepL.

Emerging technologies linked to AI

Technology	Description
Internet of Things (IoT)	Everyday objects with sensors connected to the internet
Robotics	Machines that perform physical tasks, often guided by AI
Big Data analytics	Extracting patterns from massive datasets
Cloud computing	Computing power and storage on demand over the internet
Autonomous vehicles	Self-driving cars, drones, ships
Blockchain	Decentralised, tamper-resistant record keeping
Virtual / Augmented Reality	Immersive digital experiences

KEY POINTS

The IoT–Cloud–AI pipeline: IoT sensors collect data → the cloud stores it → AI analyses it → people receive recommendations or decisions.

Benefits and concerns

Benefits of AI	Concerns and limitations of AI
Speed and efficiency	Job losses through automation
Available 24/7	Bias learned from unfair training data
Fewer errors in repetitive tasks	"Black box": decisions are hard to explain
Handles huge datasets	Privacy risks from personal data
Better prediction and diagnosis	High development cost
Long-term cost savings	Misuse: deepfakes, phishing, cyberattacks

Example: Describe how IoT, the cloud and AI could help a cocoa farmer in the South West Region.

Soil-moisture and temperature sensors (IoT) send readings every hour to a cloud server.

An AI model analyses the readings with weather forecasts and past harvests.

The farmer receives an SMS telling him when to water, spray against black pod disease or harvest.

Practice Questions: Lesson 2

1. Give two everyday examples of AI and say what each does.
2. State two benefits and two concerns of AI.

Answers: *1. E.g. face unlock (recognises the owner's face); spam filter (sorts junk email) 2. Benefits: speed, 24/7 work, fewer errors; concerns: job losses, bias, privacy, black-box decisions*

Lesson 3: AI techniques and Machine Learning

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ distinguish machine learning from traditional programming
- ✔ describe the three types of machine learning and key AI techniques

KEY POINTS

Machine Learning (ML) is a subset of AI in which a system learns patterns from data instead of being explicitly programmed with fixed rules.

Traditional programming	Machine learning
The programmer writes the rules	The system learns the rules from examples
Input + program → output	Input + expected output (training data) → model
Rules are clear and easy to edit	Learned rules are often hard to explain
Best for well-defined problems	Best for pattern-recognition problems

Types of machine learning

Type	Data used	Goal	Example
Supervised	Labelled (input + correct answer)	Predict the answer for new inputs	Spam filter, house-price prediction
Unsupervised	Unlabelled	Find hidden groups or patterns	Grouping customers by buying habits
Reinforcement	Rewards and penalties from interaction	Learn the best actions	Game-playing AI, a robot learning to walk

- Supervised learning does **classification** (predict a category: spam / not spam) or **regression** (predict a number: a price, a temperature).
- Unsupervised learning does **clustering** (grouping similar items) or **association** ("people who buy bread also buy butter").

Key AI techniques

Technique	Description	Example use
Neural networks	Layers of connected "neurons" loosely inspired by the brain	Pattern recognition
Deep learning	Neural networks with many layers	Image recognition, speech
Natural Language Processing (NLP)	Understanding and producing human language	Chatbots, translation
Computer vision	Interpreting images and video	Face recognition, self-driving cars
Expert systems	IF–THEN rules from human experts: knowledge base + inference engine + user interface	Medical diagnosis in a narrow field
Genetic algorithms	Imitate evolution to improve solutions	Timetables, design optimisation

Example: Which type of machine learning fits each task? (a) Predicting tomorrow's maize price from past prices. (b) Splitting mobile-money users into groups with similar habits, with no groups given in advance.

(a) Supervised learning (regression): past prices are labelled examples of the number to predict.

(b) Unsupervised learning (clustering): the data has no labels; the system finds the groups itself.

Practice Questions: Lesson 3

1. Which type of machine learning trains a robot by rewards and penalties?
2. Name the three parts of an expert system.

Answers: *1. Reinforcement learning 2. Knowledge base, inference engine, user interface*

Lesson 4: Developing AI Systems and the importance of data in AI systems – Part 1

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ outline the stages in developing an AI system
- ✔ explain the purpose of training and test data

Stages in developing an AI system

1. **Define the problem:** what task must the AI do, and what counts as success?
2. **Collect data:** gather enough relevant data.
3. **Prepare (clean) the data:** remove errors, duplicates, outliers and irrelevant records.
4. **Choose and train a model:** select an algorithm and let it learn from the **training set**.
5. **Test (validate) the model:** measure its accuracy on a **test set** it has never seen.
6. **Deploy:** put the system into real use.
7. **Monitor and maintain:** retrain with new data, fix errors and watch for falling accuracy.

KEY POINTS

The data is usually split into a **training set** (often about 80%) used to learn, and a **test set** (about 20%) kept back to check accuracy on unseen examples.

Example: A model correctly classifies 170 of the 200 emails in its test set. Find its accuracy.

Accuracy = correct ÷ total × 100 = 170 ÷ 200 × 100 = 85%

Practice Questions: Lesson 4

1. Why must a model be tested on data it has never seen?
2. A model is correct on 156 of 200 test cases. Find its accuracy.

Answers: *1. To check it works on new data and has not simply memorised the training examples 2. 78%*

Lesson 5: Developing AI Systems and the importance of data in AI systems – Part 2

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ explain why data quality, quantity and variety matter in AI
- ✔ explain bias in AI and the principles of data privacy

KEY POINTS

Data is the "fuel" of AI: a model is only as good as the data used to train it. **GIGO: Garbage In, Garbage Out.**

Property	Why it matters
Quantity	More examples usually give better accuracy (up to a point)
Quality	Accurate, relevant data avoids errors
Variety (representativeness)	Diverse data helps the model work on new situations and avoids bias

Bias in AI

If the training data reflects human bias, the AI learns and repeats it.

Example: Explain how a recruitment AI could become unfair to women.

It is trained on ten years of hiring records from a company that mostly hired men.

It learns that "male" profiles are linked with being hired and starts rejecting female applicants: not because it was told to, but because it copied a biased history.

Fix: use balanced, representative data, remove irrelevant attributes such as gender, and test outcomes for fairness.

Data privacy and ethics

- Collect only the data you need (**data minimisation**).
- Keep it secure (encryption, access control).
- Get **consent** where personal data is involved.
- Let people see and delete their data.
- Be transparent about how data is used, and do not use it for other purposes.

Practice Questions: Lesson 5

1. What does GIGO stand for and what does it mean for AI?
2. State three principles for handling personal data ethically.

Answers: *1. Garbage In, Garbage Out: poor training data gives poor AI decisions 2. Collect only what is needed; keep it secure; get consent; be transparent; allow access and deletion*

Lesson 6: Data collection and Data analysis

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ describe methods of data collection
- ✔ outline the data analysis process and common tools

Data collection methods

Method	Type	Example
Surveys and questionnaires	Primary	Asking students about their study habits
Interviews	Primary	One-to-one questions with farmers
Sensors (IoT)	Primary	Automatic temperature readings
Observation	Primary	Counting vehicles at a junction
Web forms	Primary	Online registration form
Databases, published reports, open government data	Secondary	INS or WHO statistics
Web scraping	Secondary	Automatically collecting prices from websites

Primary data	Secondary data
Collected directly by you for your purpose	Collected earlier by someone else
More control over quality	Quality may be unknown
Often slower and more expensive	Often cheaper and faster

KEY POINTS

Good data is **accurate, complete, relevant, up to date** and **collected ethically**.

The data analysis process

1. **Cleaning:** remove errors, duplicates and missing values.
2. **Organising:** sort and structure the data (spreadsheets, databases).
3. **Interpreting:** use statistics, charts and graphs to find patterns and trends.
4. **Concluding:** use the results to make decisions or predictions.

Tool	Use
Spreadsheets (Excel, Google Sheets)	Formulas, pivot tables, charts
Database software (MySQL, Access)	Querying data with SQL
Python (Pandas, Matplotlib)	Advanced analysis and visualisation
Power BI, Tableau	Business dashboards

Example: The marks of five students are 12, 15, 9, 15 and 14. Find the mean, the mode and the range.

Mean = (12 + 15 + 9 + 15 + 14) ÷ 5 = 65 ÷ 5 = 13

Mode = 15 (appears twice); range = 15 − 9 = 6

Practice Questions: Lesson 6

1. Classify as primary or secondary: (a) a questionnaire you give to classmates; (b) census figures from INS.
2. The ages of six students are 16, 17, 17, 18, 16, 17. Find the mean and the mode.

Answers: *1. (a) Primary (b) Secondary 2. Mean = 101 ÷ 6 ≈ 16.8; mode = 17*

Lesson 7: Integration Activity

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ apply AI concepts, development stages and data ethics to a real situation

Lesson notes

Situation: The principal of GBHS Bamenda wants an AI system that warns teachers early when a Lower Sixth student risks failing the GCE. The school has five years of records: sequence marks, attendance, age, gender and final GCE results.

Worked examples

Example: Task 1. Say which type of machine learning fits this project and why.

Supervised learning (classification): past records are labelled with the result (pass / fail), and the model predicts the label for current students.

Example: Task 2. List, in order, the stages the developers should follow.

Define the problem → collect the records → clean them → train a model on about 80% of them → test it on the other 20% → deploy it → monitor and retrain each year.

Example: Task 3. Give two risks of bias and how to reduce them.

Using gender or age could make the system unfair to some groups: remove attributes that should not affect the decision.

If most records come from one stream or one year, predictions for others may be poor: use representative data and check accuracy for each group.

Example: Task 4. State two rules the school must follow to protect students' privacy.

Use only the data needed and keep it secure with passwords and limited access.

Inform students and parents how the data is used and do not share it with outsiders.

Practice Questions: Lesson 7

1. A bank wants AI to detect mobile-money fraud. Which type of ML suits it if past transactions are labelled "fraud" or "genuine"?
2. Give one risk if the training data comes only from city users.

Answers: *1. Supervised learning (classification) 2. The model may wrongly flag normal rural transactions: the data is not representative (bias)*

Review Exercise: Exploring AI Concepts and Emerging Technologies

Answer these mixed questions after finishing the topic. They combine ideas from several lessons, like GCE questions do.

Review Questions

1. Define artificial intelligence and give three reasons why it has become practical in recent years.
2. Distinguish between narrow AI, general AI and super AI. Which one exists today?
3. Compare traditional programming with machine learning.
4. Describe the three types of machine learning with one example each.
5. List the seven stages in developing an AI system.
6. Explain the GIGO principle and its link with bias in AI.
7. Distinguish between primary and secondary data, with two examples of each.

Answers: *1. The ability of computers to do tasks that need human intelligence; practical now because of cheap powerful computing, huge amounts of data and better algorithms 2. Narrow AI does one task; general AI would match humans at any task; super AI would exceed humans; only narrow AI exists today 3. Traditional programming: data + rules → answers; machine learning: data + answers → rules (a model) 4. Supervised (labelled data, e.g. spam filter), unsupervised (unlabelled, e.g. customer grouping), reinforcement (rewards, e.g. game-playing robot) 5. Define the problem, collect data, prepare the data, choose and train a model, test it, deploy it, monitor and maintain it 6. Garbage in, garbage out: a model is only as good as its data, so biased or wrong data produces biased or wrong results 7. Primary: collected by you for your purpose (survey, interviews); secondary: collected earlier by others (INS statistics, published reports)*

END OF FREE SAMPLE

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